

STAT 100 - Important Code for Week 6 (Simulation-Based Hypothesis Tests)

Functions for Hypothesis Tests (for Sample Proportion)

Set **seed** to save randomly-generated information over time (put this at top of r-chunk)

- `set.seed(---)`
 - `set.seed(1)`

Construct data frame of sample results for a **sample proportion**

- `DATA-FRAME <- data.frame(CAT-RESPONSE-VAR = c(rep("CAT-1", VALUE-1), rep("CAT-2", VALUE-2)))`
 - `esp <- data.frame(guess = c(rep("correct", 106), rep("incorrect", 329 - 106)))`

Generate **null distribution** for a **sample proportion**

- `NULL-DIST <- DATA-FRAME %>%`
 - `specify(response = RESPONSE-VAR, success = "CAT-OF-INTEREST") %>%`
 - `hypothesize(null = "point", p = NULL-VALUE) %>%`
 - `generate(reps = 1000, type = "draw") %>%`
 - `calculate(stat = "prop")`
- `null_dist <- esp %>%`
 - `specify(response = guess, success = "correct") %>%`
 - `hypothesize(null = "point", p = 0.25) %>%`
 - `generate(reps = 1000, type = "draw") %>%`
 - `calculate(stat = "prop")`

Compute **observed test statistic** for a **sample proportion**

- `TEST-STATISTIC <- DATA-FRAME %>%`
 - `specify(response = RESPONSE-VAR, success = "CAT-OF-INTEREST") %>%`
 - `calculate(stat = "prop")`
- `test_stat <- esp %>%`
 - `specify(response = guess, success = "correct") %>%`
 - `calculate(stat = "prop")`

Compute **p-value**

- `P-VALUE <- NULL-DIST %>%`
 - `get_p_value(obs_stat = TEST-STAT, direction = "DIRECTION-OF-TEST")`
- `p_value <- null_dist %>%`
 - `get_p_value(obs_stat = test_stat, direction = "greater")`

Functions for Computing Power (for Sample Proportion)

Step #1: Construct a dataframe and generate a **null distribution**

```
- DATA-FRAME <- data.frame(CAT-RESPONSE-VAR = c(rep("CAT-1", VALUE-1),  
  rep("CAT-2", VALUE-2)))  
  - dat <- data.frame(at_bats = c(rep("hit", 80), rep("miss", 20)))  
- NULL-DIST <- DATA-FRAME %>%  
  specify(response = RESPONSE-VAR, success = "CAT-OF-INTEREST") %>%  
  hypothesize(null = "point", p = NULL-VALUE) %>%  
  generate(reps = 1000, type = "draw") %>%  
  calculate(stat = "prop")  
- null <- dat %>%  
  specify(response = at_bats, success = "hit") %>%  
  hypothesize(null = "point", p = 0.25) %>%  
  generate(reps = 1000, type = "draw") %>%  
  calculate(stat = "prop")
```

Step #2: Determine the **critical value(s)**

```
- quantile(NULL-DIST$NULL-VALUE-IN-NULL-DIST, ONE-MINUS-ALPHA)  
  - quantile(null$stat, 0.95)
```

Step #3: Generate an **alternative distribution**

```
- ALT-DIST <- DATA-FRAME %>%  
  specify(response = RESPONSE-VAR, success = "CAT-OF-INTEREST") %>%  
  hypothesize(null = "point", p = ALT-VALUE) %>%  
  generate(reps = 1000, type = "draw") %>%  
  calculate(stat = "prop")  
- alt <- dat %>%  
  specify(response = at_bats, success = "hit") %>%  
  hypothesize(null = "point", p = 0.33) %>%  
  generate(reps = 1000, type = "draw") %>%  
  calculate(stat = "prop")
```

Step #4: Find probability of **critical value(s)** (or more extreme) under **alternative distribution**

```
- ALT-DIST %>%  
  summarize(power = mean(ALT-VALUE-IN-ALT-DIST >  
    quantile(NULL-DIST$ALT-VALUE-IN-ALT-DIST, 0.95)))  
  - alt %>%  
    summarize(power = mean(stat > quantile(null$stat, 0.95)))
```